



PREDICTING YOUTUBE VIDEO VIRALITY USING CONTENT, CREATOR, AND ENGAGEMENT ATTRIBUTES

An end-to-end ML pipeline for actionable virality prediction

COURSE & TERM

MSDS_420 & Team 3

TEAM 3

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PROBLEM STATEMENT & RESEARCH QUESTIONS

🕒 BUSINESS CONTEXT

- ▶ **Problem:** Identifying viral potential early enough to optimize promotion and content strategy.
- ▶ **Unit of Analysis:** Individual YouTube videos (one video per row).
- ▶ **Target Variable:** $\log_{1p}(\text{view_count})$ to stabilize extreme variance.

❓ RESEARCH QUESTIONS

- ▶ Which features (creator vs. content) best predict video views?
- ▶ Are creator-level features more influential than content-level features?
- ▶ Which model class performs best on this tabular dataset?

Leakage Policy: No post-upload engagement (likes/comments) used for prediction.

LITERATURE REVIEW: STRUCTURAL DRIVERS OF VIRALITY

👥 CREATOR SOCIAL CAPITAL

- ▶ **Subscriber Base:** Strongly predicts views, likes, and comments (Hussain et al. 2024).
- ▶ **Exposure Networks:** Larger channels benefit from accumulated audience reach, amplifying initial visibility.
- ▶ **Platform Inequality:** YouTube follows a "winner-take-most" distribution where top creators capture disproportionate attention.

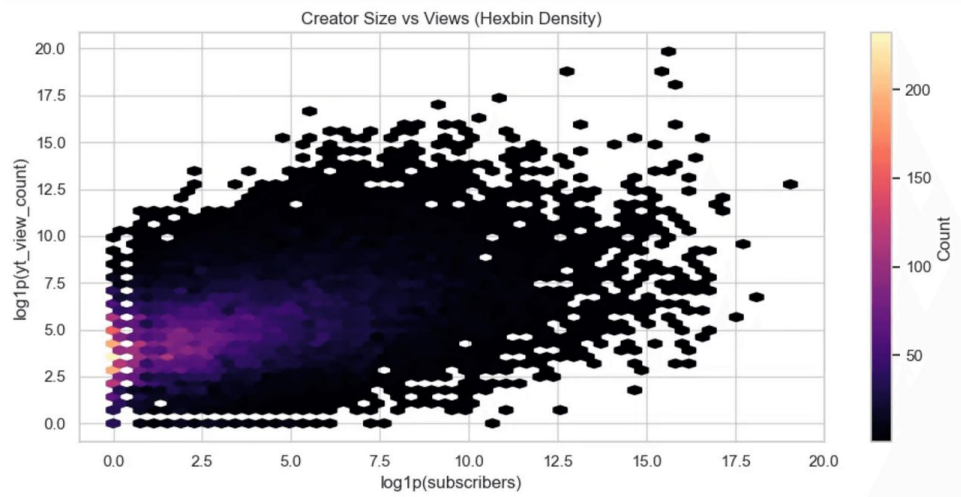
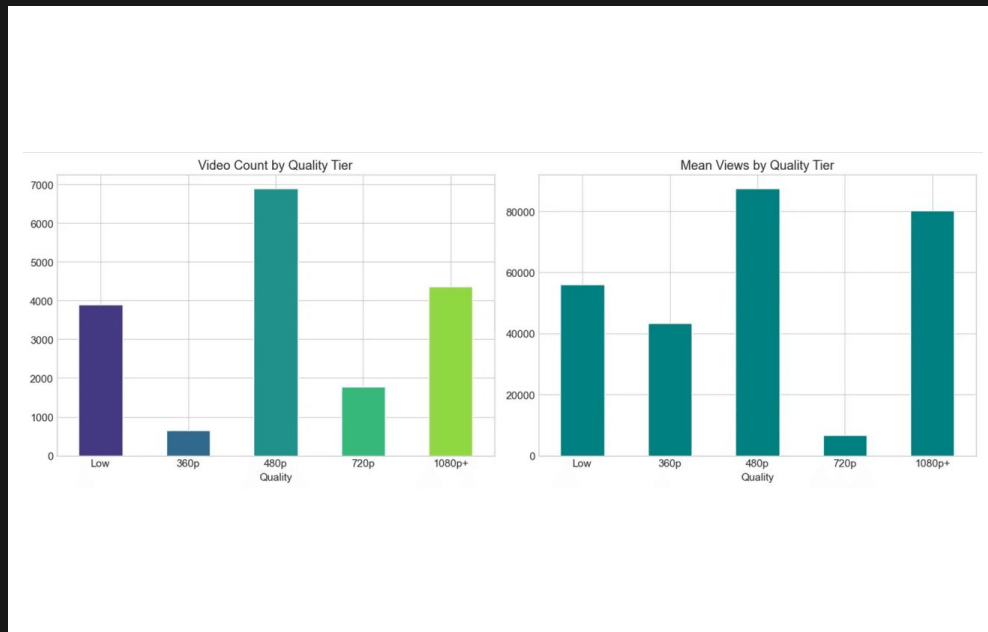


Figure 1: Hexbin density plot showing the strong positive correlation between creator size (subscribers) and video views.

LITERATURE REVIEW: VISUAL ATTENTION & PLATFORM DYNAMICS

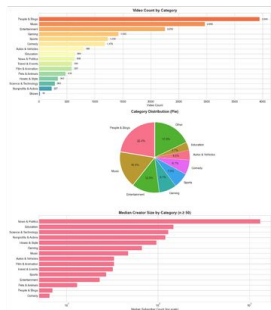
👁️ VISUAL & NONLINEAR DRIVERS

- ▶ **Thumbnail Salience:** Brightness and colorfulness act as heuristics for content quality, influencing click-through (Koh & Cui 2022).
- ▶ **Nonlinear Dynamics:** Virality emerges from complex interactions, not linear additive effects (Peng & Bainbridge 2026).
- ▶ **Duration Myth:** No consistent linear relationship between video length and engagement (Konukoğlu & Özman 2025).



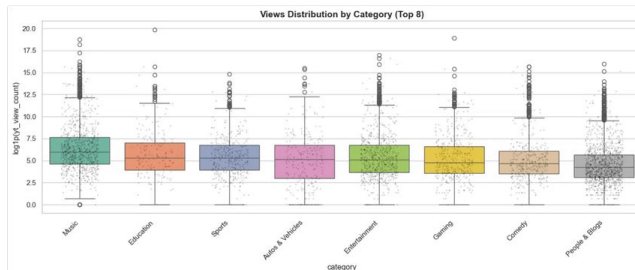
EXPLORATORY DATA ANALYSIS: CATEGORY & DISTRIBUTION

CATEGORY DISTRIBUTION



-  **Dominant Categories:** People & Blogs, Music, and Entertainment account for >50% of the dataset.
-  **Creator Size:** News & Politics channels have the highest median subscriber count.

VIEWS DISTRIBUTION BY CATEGORY (LOG-SCALE)



EXPLORATORY DATA ANALYSIS: CORRELATION & FEATURES

STRUCTURAL HIERARCHY

- ▶ **Channel Dominance:** Subscriber count and historical views show stronger correlation with video views than content features.
- ▶ **Engagement Clustering:** Views, likes, and comments are highly collinear, validating the use of views as the primary target.
- ▶ **Weak Content Signal:** Duration and frame rate show negligible linear correlation with performance.

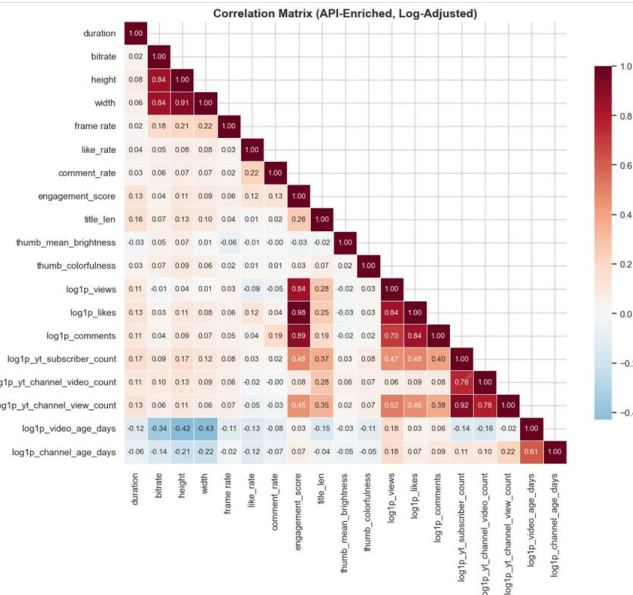
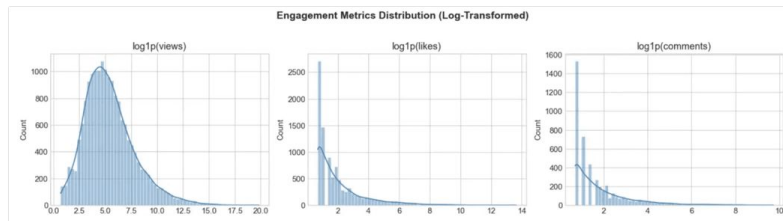


Figure 3: Correlation matrix highlighting the strong relationship between channel-level variables and engagement metrics.

EDA: ENGAGEMENT PATTERNS & DISTRIBUTION

LOG-TRANSFORMED ENGAGEMENT DISTRIBUTIONS



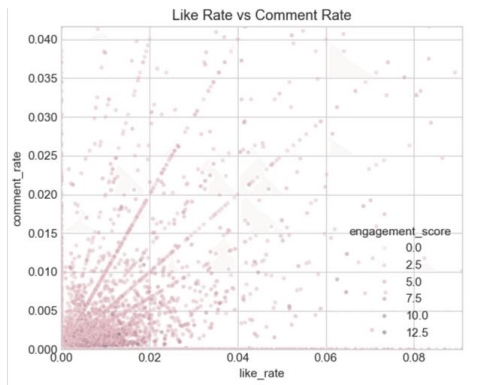
RIGHT-SKEWED DISTRIBUTIONS

After log1p transformation, views approximate a normal distribution (peak ~5.0), while likes and comments remain right-skewed with zero-inflation.

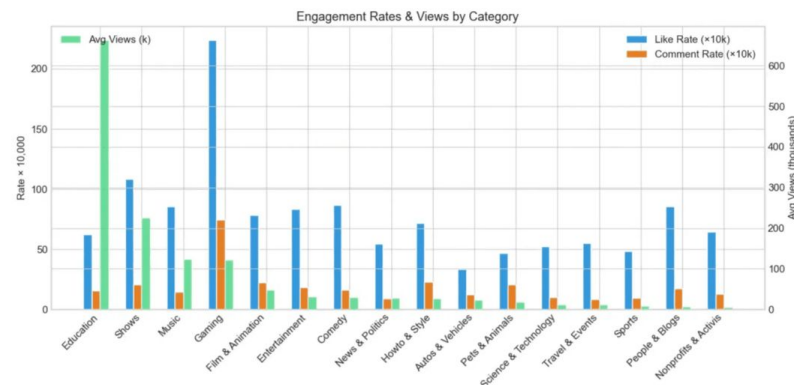
IMPLICATION

The skewness justifies using log-transformed views as the regression target, reducing the influence of viral outliers on model training.

LIKE RATE VS COMMENT RATE



ENGAGEMENT RATES & VIEWS BY CATEGORY



FEATURE ENGINEERING & PREPROCESSING



CREATOR FEATURES

- Subscriber count
- Total channel views
- Historical avg views/video



TEMPORAL FEATURES

- Video age (days since upload)
- Channel age (account maturity)
- Upload frequency metrics



CONTENT PROXIES

- Title & description length
- Thumbnail brightness
- Thumbnail colorfulness

PREPROCESSING PIPELINE

Imputation

Median (Numeric) / Mode (Cat)



Encoding

One-Hot for Categories



Scaling

StandardScaler for NN/KNN



Leakage Check

Strict Feature Isolation

METHODOLOGY & EXPERIMENTAL SETUP



EXPERIMENTAL DESIGN

- ▶ **Data Split:** 80% Training / 20% Testing with a fixed random seed for reproducibility.
- ▶ **Hyperparameter Tuning:** 5-fold GridSearchCV applied to the Random Forest model.
- ▶ **Selection Criterion:** Best generalization on test RMSE + model interpretability.



MODEL EVALUATION

- ▶ **Models Evaluated:**
 - Elastic Net BASILINE
 - KNN Regressor
 - Random Forest Regressor
 - MLP Regressor NEURAL NET
- ▶ **Performance Metrics:** RMSE, MAE, and R^2 (calculated on log-transformed views).

FINDINGS & MODEL COMPARISON

PERFORMANCE METRICS

MODEL	RMSE	MAE	R ²
Elastic Net	1.852	1.420	0.451
KNN Regressor	1.710	1.305	0.512
Random Forest (Tuned)	1.507	1.120	0.621
MLP Regressor	1.685	1.290	0.530

BEST MODEL: TUNED RANDOM FOREST

KEY INTERPRETATIONS

- ▶ **Social Capital Dominance:** Creator-level features (subscribers/history) are the strongest predictors.
- ▶ **Visual Signal:** Thumbnail variables add secondary but statistically meaningful signal.
- ▶ **Nonlinear Advantage:** Tree-based models outperform linear ones due to complex feature interactions.
- ▶ Practical, decision-supportive pipeline for early promotion strategy.

LIMITATIONS & TEAM CONTRIBUTIONS

LIMITATIONS

- ▶ **Sampling Bias:** Data may reflect specific seasonal trends or temporal anomalies.
- ▶ **Semantic Depth:** Metadata-focused modeling misses narrative nuance and video content quality.
- ▶ **Predictive Scope:** Analysis identifies correlations, not causal mechanisms.
- ▶ **Data Gaps:** Missingness in enriched fields (e.g., country) affects localized predictions.

TEAM ROLES

BUSINESS FRAMING

Research questions, literature review, and value proposition definition.

EDA & PREPROCESSING

Data cleaning, visualization, feature engineering, and pipeline design.

MODELING & INTEGRATION

Model selection, hyperparameter tuning, and deployment strategy.

DEPLOYMENT STRATEGY & AUTOMATION



AUTOMATION FLOW

- ▶ End-to-end pipeline from raw data to actionable reporting.



- ▶ Optional API enrichment for real-time channel metadata.



DEPLOYMENT OPTIONS

- ▶ **Batch Scoring:** Scheduled retraining and inference for portfolio tracking.
- ▶ **API Serving:** On-demand prediction for individual video uploads.



REPRODUCIBILITY

- ▶ Fixed random seeds and versioned dependencies.
- ▶ Environment-based credential management for secure API access.
- ▶ Logged experiment tracking for model auditability.

MANAGEMENT RECOMMENDATIONS

STRATEGIC ACTIONS

- ▶ **Operationalize Scoring:** Implement early-stage prediction to prioritize promotion budgets.
- ▶ **Thumbnail Optimization:** Systematize A/B testing based on brightness and color diversity.
- ▶ **Creator Growth:** Prioritize long-term partnerships with creators showing high social capital growth.

TECHNICAL ROADMAP

- ▶ **Advanced Modeling:** Evaluate XGBoost, LightGBM, and CatBoost for marginal gains.
- ▶ **Contextual Data:** Integrate Google Trends and external social signals.
- ▶ **Risk Management:** Implement prediction intervals and uncertainty reporting.

FROM PREDICTIVE INSIGHTS TO ACTIONABLE COMPETITIVE ADVANTAGE

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